

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 6: Keeping Cool and Staying Warm: Thermoregulation in the Dog, the Lizard and Us

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	C	C	A	B	C

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on why the dog pants while the lizard stays still on a hot rock

Q1. On a hot afternoon in Kericho, a dog pants rapidly with its tongue hanging out. How does panting help the dog control its body temperature?

- A. The dog sweats through its tongue, just as humans sweat through the skin
- B. Panting warms the dog's blood so it matches the air temperature
- C. Panting makes the dog's body produce more heat so it can adapt to the weather
- D. Water evaporates from the tongue and airways and carries heat away from the body**

Correct answer: D. Panting moves air over the wet tongue and airways. Evaporating water takes heat from the body, so the dog cools. Dogs do not sweat through the tongue.

USE AFTER: OBSERVE PHASE — Observe, Station 1: Diagrams, annotating vasodilation and vasoconstriction on the mammal skin and hypothalamus chart

Q2. Blood in a mammal becomes too hot. Which response in the skin helps to lower body temperature?

- A. Vasodilation, which brings more blood close to the skin surface so heat is lost**
- B. Vasoconstriction, which reduces blood flow near the skin surface
- C. Shivering, which makes the muscles contract rapidly
- D. Hair erection, which traps a layer of air next to the skin

Correct answer: A. Vasodilation widens skin arterioles so more blood flows near the surface and heat is lost. Vasoconstriction, shivering and hair erection all conserve or produce heat.

USE AFTER: OBSERVE PHASE — Observe, Station 3: Field observation notes on the agama lizard at 9 am, 11 am and 2 pm

Q3. A field note says: 'Agama lizard on a rock at 9 am and 11 am. At 2 pm, when the rock is very hot, it moves into the shade.' What is the best explanation for this behaviour?

- A. The lizard is lazy and rests when it is tired
- B. The lizard produces its own heat and moves to the shade to burn it off
- C. The lizard cannot control its temperature internally, so it moves to avoid overheating**
- D. The lizard's hypothalamus makes it sweat in the shade

Correct answer: C. The lizard is an ectotherm and depends on the environment for heat. It basks to warm up and moves into shade when it is too hot. The timing of the move is evidence against laziness.

USE AFTER: EXPLAIN PHASE — Explain: building the flowchart of the negative feedback loop for body temperature

Q4. Body temperature control is called a negative feedback loop. Why is it called 'negative'?

- A. The response makes the body temperature change even more in the same direction
- B. Negative feedback is harmful to the body and must be avoided
- C. The response reverses the change that triggered it and brings the temperature back to normal**
- D. The control centre switches off and stops responding to temperature changes

Correct answer: C. In negative feedback, the response opposes the original change. For example, sweating cools a hot body, which then removes the stimulus.

USE AFTER: EXPLAIN PHASE — Explain: the reverse loop for cold (vasoconstriction, shivering, hair erection)

Q5. A goat is standing in the cold early morning and its body temperature begins to fall. Which set of responses helps it to conserve and produce heat?

- A. Vasoconstriction, shivering and hair erection**
- B. Vasoconstriction, panting and hair erection
- C. Vasodilation, sweating and panting
- D. Vasodilation, shivering and sweating

Correct answer: A. In the cold loop, vasoconstriction reduces heat loss from the skin, shivering produces heat in the muscles, and hair erection traps insulating air. Vasodilation, sweating and panting are cooling responses.

USE AFTER: EXPLAIN PHASE — Explain: matching local examples (burrowing, hibernation, aestivation) to the problem each solves

Q6. Some frogs and lungfish bury themselves in mud during the long dry season at the shores of Lake Victoria and become inactive. What problem does this response (aestivation) help them survive?

- A. Too much water in their bodies
- B. Prolonged heat and drought**
- C. Prolonged cold weather
- D. A lack of food caused by too many predators

Correct answer: B. Aestivation is a long period of inactivity that helps an animal survive prolonged heat and drought. Hibernation is the response to prolonged cold.

USE AFTER: MODEL BUILDING PHASE — Model: adding the thermoregulation loop to the cumulative model and linking it to water balance (the dog's water cost of panting)

Q7. A dog pants for a long time on a hot day. Besides cooling the dog, what is another effect of the panting?

- A. The dog's body stops using the hypothalamus because it is already cooling
- B. The dog's kidneys make more dilute urine to remove the extra water
- C. The dog loses water by evaporation, so its body must conserve water and the dog needs to drink**
- D. The dog gains water from the air it breathes in

Correct answer: C. Panting cools by evaporating water from the tongue and airways, so the dog loses body water. This links temperature control to water balance, because the kidneys must conserve water and the dog must drink.